Multiple Linear Regression & Predictive Modeling

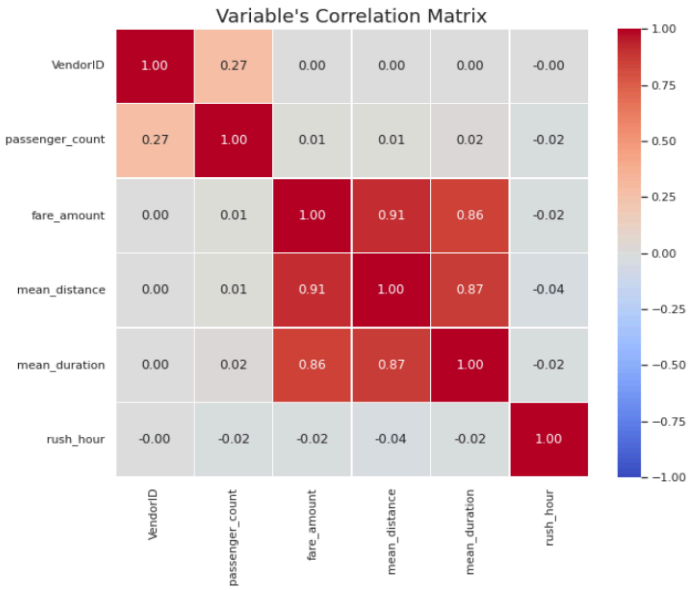
Executive Summary Report: NYC TLC Data Project

Prepared by Automatidata for New York City Taxi & Limousine Commission

Overview

The overall purpose of this stage of the project is to construct a Multiple Linear Regression (MLR) model to predict taxi fare amounts prior to a ride taking place. By leveraging historical trip records and feature engineering techniques—such as calculating mean trip distance and mean duration grouped by pickup and drop-off locations—Automatidata provides NYC TLC with a robust predictive tool to estimate fare structures, optimize driver earnings, and implement automated rate rules.

Problem



During exploratory data analysis and initial feature evaluation, two main structural challenges were identified:

- **Multicollinearity and Variable Redundancy:** Trip distance and trip duration show a high positive correlation with fare-amount, as well as a strong correlation between each other. Using both directly as raw predictors introduces collinearity risks into linear modeling.
- **Deterministic Flat-Rate Anomalies (JFK Trips):** Trips originating or terminating at Location 132 consistently recorded a flat fare of $52.00 paired with RatecodeID 2 (JFK Airport). These fixed-rate scenarios require dedicated handling to avoid skewing continuous regression predictions.

Key Insights (MLR & Feature Engineering)

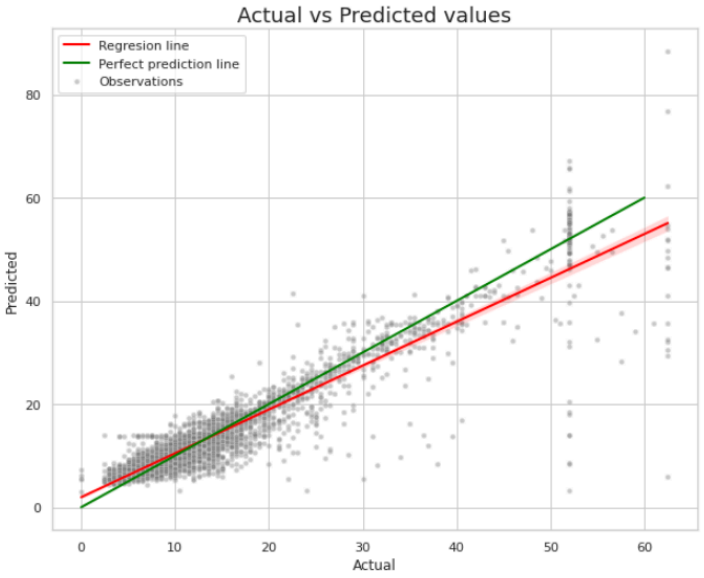
Feature Engineering & Modeling Findings:

- **Mean Distance & Duration Aggregations:** Mean trip distance and mean duration were calculated by grouping pickup and drop-off location pairs (PULocationID / DOLocationID). This captures spatial traffic patterns and stabilizes predictive signals across similar routes.
- **JFK Airport Flat-Fare Rules (RatecodeID 2 / Location 132):** Analysis of $52.00 fares revealed consistent toll charges ($5.76 / $5.54) and confirmed 2017 official NYC regulations for JFK-Manhattan flat rates. Because RatecodeID 2 follows a deterministic rule, these fares can be imputed post-prediction to guarantee 100% accuracy.
- **Outlier Removal & Cleaning:** Extreme outliers, including records with 0.0 miles but positive fares, were filtered out to prevent linear coefficient distortion.

Details & Model Evaluation

Quantitative evaluation metrics for the fitted Multiple Linear Regression model on test data:

Evaluation Metric	Test Score / Value	Business Interpretation
R-squared (R^2)	0.868	The model explains ~86.8% of the total variance in taxi fare amounts.
MAE (Mean Absolute Error)	$2.13	On average, fare predictions deviate by approximately $2.13 per trip.
RMSE (Root Mean Sq. Error)	$3.77	Reflects low overall prediction variance while accounting for larger residuals.



Next Steps

Key recommendations for business deployment and pipeline integration:

- **Post-Prediction JFK Imputation:** Incorporate deterministic rule-based post-processing so that trips with RatecodeID 2 (or Location 132) automatically assign a $52.00 flat fare, bypassing regression error.
- **Upstream Pre-Ride Estimation:** Deploy the model into customer app interfaces to provide transparent pre-ride fare estimates using historical route mean_distance and mean_duration.
- **Advanced Non-Linear Modeling:** Explore tree-based models (Random Forest, XGBoost) in subsequent phases to capture non-linear traffic dynamics and complex time-of-day interactions.